

Surds and simplification

The square root of $14 + 6\sqrt{5}$ is

- (1) $2 + \sqrt{5}$ (2) $3 + \sqrt{5}$
(3) $5 + \sqrt{3}$ (4) $3 + 2\sqrt{5}$

$$\sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7 + 4\sqrt{3}}}} \text{ is}$$

- | | |
|-------|-------|
| (1) 1 | (2) 2 |
| (3) 3 | (4) 8 |

$$\sqrt{4600 + \sqrt{540 + \sqrt{1280 + \sqrt{250 + \sqrt{36}}}}} ?$$

- | | |
|--------|--------|
| (a) 69 | (b) 68 |
| (c) 70 | (d) 72 |

The value of $\frac{\sqrt{0.6912} + \sqrt{0.5292}}{\sqrt{0.6912} - \sqrt{0.5292}}$ is:

- | | |
|---------|---------|
| (a) 1.5 | (b) 9 |
| (c) 15 | (d) 0.9 |

Given that $\sqrt{3} = 1.732$, the value of

$$\frac{3 + \sqrt{6}}{5\sqrt{3} - 2\sqrt{12} - \sqrt{32} + \sqrt{50}}$$
 is

- | | |
|-----------|-----------|
| (1) 4.899 | (2) 2.551 |
| (3) 1.414 | (4) 1.732 |

$$\frac{1}{\sqrt{3.25} + \sqrt{2.25}} + \frac{1}{\sqrt{4.25} + \sqrt{3.25}} + \frac{1}{\sqrt{5.25} + \sqrt{4.25}} + \frac{1}{\sqrt{6.25} + \sqrt{5.25}} \text{ is}$$

(1) 1.00 (2) 1.25
(3) 1.50 (4) 2.25

$$\frac{\sqrt{5}}{\sqrt{3} + \sqrt{2}} - \frac{3\sqrt{3}}{\sqrt{5} + \sqrt{2}} + \frac{2\sqrt{2}}{\sqrt{5} + \sqrt{3}}$$

is equal to :

- (1) 0 (2) $2\sqrt{15}$
(3) $2\sqrt{10}$ (4) $2\sqrt{6}$

$$\frac{2+\sqrt{3}}{2-\sqrt{3}} + \frac{2-\sqrt{3}}{2+\sqrt{3}} + \frac{\sqrt{3}+1}{\sqrt{3}-1} \text{ is}$$

$$(1) \ 16+\sqrt{3} \quad (2) \ 4-\sqrt{3}$$

$$(3) \ 2-\sqrt{3} \quad (4) \ 2+\sqrt{3}$$

$$\sqrt{\frac{(\sqrt{12}-\sqrt{8})(\sqrt{3}+\sqrt{2})}{5+\sqrt{24}}} \text{ is :}$$

$$(1) \ \sqrt{6}-\sqrt{2} \quad (2) \ \sqrt{6}+\sqrt{2}$$

$$(3) \ \sqrt{6}-2 \quad (4) \ 2-\sqrt{6}$$

$$\left[\frac{1}{\sqrt{2} + \sqrt{3} - \sqrt{5}} + \frac{1}{\sqrt{2} - \sqrt{3} - \sqrt{5}} \right]$$

in simplified form equals to :

(1) 1 (2) $\sqrt{2}$

(3) $\frac{1}{\sqrt{2}}$ (4) 0

$$\left(\frac{2}{\sqrt{6} + 2} + \frac{1}{\sqrt{7} + \sqrt{6}} + \frac{1}{\sqrt{8} - \sqrt{7}} + 2 - 2\sqrt{2} \right)$$

is equal to

(1) 0 (2) $2\sqrt{2}$

(3) $\sqrt{6}$ (4) $2\sqrt{7}$

$$\frac{2\sqrt{10}}{\sqrt{5} + \sqrt{2} - \sqrt{7}} - \sqrt{\frac{\sqrt{5} - 2}{\sqrt{5} + 2}} - \frac{3}{\sqrt{7} - 2} \text{ is:}$$

SSC CGL Tier II (13/09/2019)

- (a) $2 + \sqrt{2}$ (b) $2\sqrt{5}$ (c) $\sqrt{2}$ (d) $\sqrt{7}$

If $\sqrt{10 - 2\sqrt{21}} + \sqrt{8 + 2\sqrt{15}} = \sqrt{a} + \sqrt{b}$, where a and b are positive integers, then the value of $\sqrt{ab}$ is closest to :

- (a) 7.2 (b) 4.6
(c) 5.9 (d) 6.8

$$\text{If } \frac{\sqrt{38 - 5\sqrt{3}}}{\sqrt{26 + 7\sqrt{3}}} = \frac{a + b\sqrt{3}}{23}, \text{ } b > 0,$$

then the value of $(b - a)$ is :

- (a) 7 (b) 18 (c) 29 (d) 11

The value of $5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - \frac{23}{\sqrt{2} + \sqrt{3} + \sqrt{6}}$

is:

- (a) 0 (b) 10
(c) 12 (d) 16

$\frac{12}{3 + \sqrt{5} + 2\sqrt{2}}$ is equal to

(1) $1 - \sqrt{5} + \sqrt{2} + \sqrt{10}$

(2) $1 + \sqrt{5} + \sqrt{2} - \sqrt{10}$

(3) $1 + \sqrt{5} - \sqrt{2} + \sqrt{10}$

(4) $1 - \sqrt{5} - \sqrt{2} + \sqrt{10}$

The expression $\sqrt{10 + 2(\sqrt{6} - \sqrt{15} - \sqrt{10})}$ is equal to :

(a) $\sqrt{3} - \sqrt{2} - \sqrt{5}$

(b) $\sqrt{2} - \sqrt{3} - \sqrt{5}$

(c) $\sqrt{3} - \sqrt{2} + \sqrt{5}$

(d) $\sqrt{3} + \sqrt{2} - \sqrt{5}$

$$\frac{15(\sqrt{10} + \sqrt{5})}{\sqrt{10} + \sqrt{20} + \sqrt{40} - \sqrt{5} - \sqrt{80}} \text{ is equal to}$$

SSC CGL Tier II (15/10/2020)

- (a) $5 + 2\sqrt{2}$ (b) $5 - 2\sqrt{5}$
 (c) $5(3 + 2\sqrt{2})$ (d) $10(3 + 2\sqrt{5})$

If $(\sqrt{2} + \sqrt{5} - \sqrt{3}) \times k = -12$, then what will be the value of k ?

- (a) $\sqrt{2} + \sqrt{5} + \sqrt{3}$
 (b) $(\sqrt{2} + \sqrt{5} - \sqrt{3})(2 + \sqrt{5})$
 (c) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{5})$
 (d) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{10})$

If $\frac{4}{1+\sqrt{2}+\sqrt{3}} = a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$, where a,

b, c, d are natural numbers, then the value of a + b + c + d is:

(a) 4

(b) 1

(c) 2

(d) 0

$$\begin{aligned} & \frac{1}{4-\sqrt{15}} - \frac{1}{\sqrt{15}-\sqrt{14}} + \frac{1}{\sqrt{14}-\sqrt{13}} - \\ & \frac{1}{\sqrt{13}-\sqrt{12}} + \frac{1}{\sqrt{12}-\sqrt{11}} - \frac{1}{\sqrt{11}-\sqrt{10}} \\ & + \frac{1}{\sqrt{10}-3} - \frac{1}{3-\sqrt{8}} \text{ is:} \end{aligned}$$

SSC CGL Tier II (29/01/2022)

(a) $4 - 2\sqrt{2}$

(b) $2 + 2\sqrt{2}$

(c) $4 + 2\sqrt{2}$

(d) $2 - 2\sqrt{2}$

$$\frac{1}{\sqrt{9}-\sqrt{8}} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-\sqrt{4}}$$

is equal to

- | | |
|-------|-------|
| (1) 5 | (2) 1 |
| (3) 3 | (4) 0 |

$$\frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-2} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{3-\sqrt{8}} \text{ is}$$

- | | |
|-------|-------|
| (1) 7 | (2) 0 |
| (3) 1 | (4) 5 |

The value of $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}}$
 $+ \frac{1}{\sqrt{4}+\sqrt{3}} + \dots + \frac{1}{\sqrt{100}+\sqrt{99}}$ is

(1) 1 (2) 9
 (3) $\sqrt{99}$ (4) $\sqrt{99} - 1$

$\frac{1}{\sqrt{100}-\sqrt{99}} - \frac{1}{\sqrt{99}-\sqrt{98}} +$
 $\frac{1}{\sqrt{98}-\sqrt{97}} - \frac{1}{\sqrt{97}-\sqrt{96}} + \dots +$
 $\frac{1}{\sqrt{2}-\sqrt{1}}$

(1) 0 (2) 9
 (3) 10 (4) 11

Thankyou

Surds and simplification

$\sqrt{3\sqrt{3\sqrt{3}\dots}}$ is equal to

(1) $\sqrt{3}$

(2) 3

(3) $2\sqrt{3}$

(4) $3\sqrt{3}$

If $x = \sqrt{4\sqrt{4\sqrt{4\sqrt{4\dots}}}}$,
 $x?$

- 2
- 4
- 6
- 8

$$\sqrt[3]{13\sqrt[3]{13\sqrt[3]{13\dots\infty}}}$$

$$\sqrt{3\sqrt{3\sqrt{3\sqrt{3}}}}$$

$$\sqrt[3]{13\sqrt[3]{13\sqrt[3]{13\sqrt[3]{13}}}}$$

$$\sqrt[4]{7 \sqrt[4]{7 \sqrt[4]{7}}}$$

$$\sqrt{7 \sqrt{7 \sqrt{7 \sqrt{7 \sqrt{7}}}}} \dots 10 \text{ terms} = ?$$

A) $7^{\frac{1023}{1022}}$

(B) $7^{\frac{1}{1023}}$

C) $7^{\frac{1}{1024}}$

(D) $7^{\frac{1023}{1024}}$

(4) 2

$\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$ is equal to

(1) 3

(2) 4

(3) 5

(4) 6

$\sqrt{7 + \sqrt{7 + \sqrt{7 + \dots \infty}}}$

A) $\frac{\sqrt{29} - 3}{2}$

B) $\frac{\sqrt{29} + 3}{2}$

C) $\frac{\sqrt{29} - 1}{2}$

D) $\frac{\sqrt{29} + 1}{2}$

$$\sqrt{17 - \sqrt{17 - \sqrt{17 - \dots - \infty}}}$$

A) $\frac{\sqrt{69} - 1}{2}$
 C) $\frac{\sqrt{69} - 3}{2}$

B) $\frac{\sqrt{69} + 3}{2}$
 D) $\frac{\sqrt{69} + 5}{2}$

$$\sqrt{72 - \sqrt{72 - \sqrt{72 - \sqrt{72 - \dots - \infty}}}} = ?$$

(A) 8

(B) 6

(C) 4

(D) 2

6
7
8
9

$$\sqrt{35 + 2\sqrt{35 + 2\sqrt{35 + 2\ldots\infty}}} = ?$$

- (A) 7 (B) 5
 (C) 9 (D) 3

$$\sqrt{54 - 3\sqrt{54 - 3\sqrt{54 - 3\ldots\infty}}} = ?$$

(A) 6

(B) 8

(C) 4

(D) 2

$$\sqrt{3 - \sqrt{3 + \sqrt{3 - \sqrt{3 + \ldots\infty}}}} = ?$$

(A) 0

(B) 1

(C) 2

(D) 3

(D) $2\sqrt{13}$

D) 12

A) 2 C) 4
B) 3 D) 8

यदि $\sqrt{x}\sqrt{x}\sqrt{x}\sqrt{x}\dots\infty = 4$ हो, तो $x = ?$

(A) 3 (B) 1
(C) 4 (D) 2

$$\sqrt{1 + 2\sqrt{1 + 3\sqrt{1 + 4\sqrt{1 + 5\ldots\infty}}}} = ?$$

(A) 1

(B) 2

(C) 4

(D) 3

Thankyou

Comparison

Which of the following number is the least?

$$(0.5)^2, \sqrt{0.49}, \sqrt[3]{0.008}, 0.23$$

(1) $(0.5)^2$ (2) $\sqrt{0.49}$

(3) $\sqrt[3]{0.008}$ (4) 0.23

Which of the following statement(s) is/are TRUE ?

I.

$$\sqrt{(64)} + \sqrt{(0.0064)} + \sqrt{(0.81)} + \sqrt{(0.0081)} = 9.07$$

II. $\sqrt{(0.010201)} + \sqrt{(98.01)} + \sqrt{(0.25)} = 11.51$

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither I nor II

Which of the following statement(s) is/are TRUE ?

I. $(0.7)^2 + (0.07)^2 + (11.1)^2 > 123.8$

II. $(1.12)^2 + (10.3)^2 + (1.05)^2 > 108.3$

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither I nor II

Arrange the following in descending order : $\sqrt[3]{4}$, $\sqrt{2}$, $\sqrt[6]{3}$, $\sqrt[4]{5}$

(1) $\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$

(2) $\sqrt[4]{5} > \sqrt[3]{4} > \sqrt[6]{3} > \sqrt{2}$

(3) $\sqrt{2} > \sqrt[6]{3} > \sqrt[3]{4} > \sqrt[4]{5}$

(4) $\sqrt[6]{3} > \sqrt[4]{5} > \sqrt[3]{4} > \sqrt{2}$

The greatest of the numbers $\sqrt[2]{8}$,

$\sqrt[4]{13}$, $\sqrt[5]{16}$, $\sqrt[10]{41}$ is:

(1) $\sqrt[4]{13}$ (2) $\sqrt[5]{16}$

(3) $\sqrt[10]{41}$ (4) $\sqrt[2]{8}$

The greatest among the numbers

$3\sqrt{2}$, $3\sqrt{7}$, $6\sqrt{5}$, $2\sqrt{20}$ is

(1) $3\sqrt{2}$

(2) $3\sqrt{7}$

(3) $6\sqrt{5}$

(4) $2\sqrt{20}$

The greatest number among 2^{60} , 3^{48} , 4^{36} and 5^{24} is

(1) 2^{60}

(2) 3^{48}

(3) 4^{36}

(4) 5^{24}

The greatest number among 3^{50} , 4^{40} , 5^{30} and 6^{20} is

- | | |
|--------------|--------------|
| (1) 3^{50} | (2) 4^{40} |
| (3) 5^{30} | (4) 6^{20} |

Which of the following statement(s) is/are TRUE ?

I. $33^3 > 3^{33}$

II. $333 > (3^3)^3$

- | | |
|-------------------|----------------------|
| (a) Only I | (b) Only II |
| (c) Both I and II | (d) Neither I nor II |

Which of the following fractions are in descending order?

निम्न में से कौन सी भिन्न अवरोही क्रम में हैं?

(A) $\frac{5}{8}, \frac{7}{12}, \frac{3}{4}, \frac{13}{16}$

(B) $\frac{7}{12}, \frac{13}{16}, \frac{3}{4}, \frac{5}{8}$

(C) $\frac{5}{8}, \frac{7}{12}, \frac{13}{16}, \frac{3}{4}$

(D) $\frac{13}{16}, \frac{3}{4}, \frac{5}{8}, \frac{7}{12}$

Which of the following fractions are in descending order?



निम्न में से कौन सी भिन्न अवरोही क्रम में हैं?

$\frac{5}{6}, \frac{3}{7}, \frac{8}{9}, \frac{3}{14}$

(A) $\frac{8}{9}, \frac{5}{6}, \frac{3}{7}, \frac{3}{14}$

(B) $\frac{8}{9}, \frac{3}{14}, \frac{3}{7}, \frac{5}{6}$

(C) $\frac{5}{6}, \frac{8}{9}, \frac{3}{7}, \frac{3}{14}$

(D) $\frac{3}{7}, \frac{8}{9}, \frac{5}{6}, \frac{3}{14}$

Which of the following is true ?

(A) $\frac{12}{43} < \frac{32}{67} < \frac{45}{81} < \frac{22}{55}$

(B) $\frac{12}{43} < \frac{22}{55} < \frac{45}{81} < \frac{32}{67}$

(C) $\frac{12}{43} < \frac{32}{67} < \frac{22}{55} < \frac{45}{81}$

(D) $\frac{12}{43} < \frac{22}{55} < \frac{32}{67} < \frac{45}{81}$

212. Which one of the following is the largest number among 2222^2 , 222^2 , 22^{222} , 2^{2222} ?

Which of the following statement(s) is/are TRUE ?

I. $\sqrt{12} > \sqrt[3]{16} > \sqrt[4]{24}$

II. $\sqrt[3]{25} > \sqrt[4]{32} > \sqrt[6]{48}$

III. $\sqrt[4]{9} > \sqrt[3]{15} > \sqrt[6]{24}$

(a) Only I and II

(b) Only I and III

(c) Only I

(d) All are true

If $A = 2^{32}$, $B = 2^{31} + 2^{30} + 2^{29} + \dots + 2^0$ and $C = 3^{15} + 3^{14} + 3^{13} + \dots + 3^0$, then which of the following option is TRUE ?

(a) $C > B > A$

(b) $C > A > B$

(c) $A > B > C$

(d) $A > C > B$

If $P = 2^{29} \times 3^{21} \times 5^8$, $Q = 2^{27} \times 3^{21} \times 5^8$, $R = 2^{26} \times 3^{22} \times 5^8$ and $S = 2^{25} \times 3^{22} \times 5^9$, then which of the following is TRUE ?

- (a) $P > S > R > Q$ (b) $S > P > R > Q$
(c) $P > R > S > Q$ (d) $S > P > Q > R$

Which is the greatest among

$$(\sqrt{19} - \sqrt{17}), (\sqrt{13} - \sqrt{11}),$$

$$(\sqrt{7} - \sqrt{5}) \text{ and } (\sqrt{5} - \sqrt{3}) ?$$

$$(1) \sqrt{19} - \sqrt{17} \quad (2) \sqrt{13} - \sqrt{11}$$

$$(3) \sqrt{7} - \sqrt{5} \quad (4) \sqrt{5} - \sqrt{3}$$

Which one among the following is the smallest?

(a) $\sqrt{301} - \sqrt{299}$

(b) $\sqrt{401} - \sqrt{399}$

(c) $\sqrt{101} - \sqrt{99}$

(d) $\sqrt{201} - \sqrt{199}$

Which of the following statement(s) is/are TRUE ?

I. $\sqrt{5} + \sqrt{5} > \sqrt{7} + \sqrt{3}$

II. $\sqrt{6} + \sqrt{7} > \sqrt{8} + \sqrt{5}$

III. $\sqrt{3} + \sqrt{9} > \sqrt{6} + \sqrt{6}$

(a) Only I

(b) Only I and II

(c) Only II and III

(d) Only I and III

The smallest of $\sqrt{8} + \sqrt{5}$,
 $\sqrt{7} + \sqrt{6}$, $\sqrt{10} + \sqrt{3}$ and
 $\sqrt{11} + \sqrt{2}$ is :

- (1) $\sqrt{8} + \sqrt{5}$ (2) $\sqrt{7} + \sqrt{6}$
(3) $\sqrt{10} + \sqrt{3}$ (4) $\sqrt{11} + \sqrt{2}$ —

Thankyou